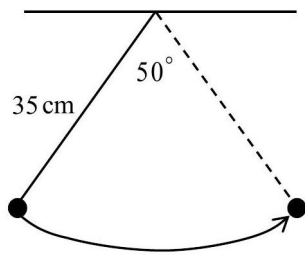


mb-pc40s-2024-jan

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

A pendulum that is 35 cm long swings through an angle of 50° . Determine the length of the arc through which the pendulum swings.



Solution

$$\begin{aligned}\theta &= 50 \left(\frac{\pi}{180} \right) \quad 1 \text{ mark for conversion} \\ &= \frac{5\pi}{18}\end{aligned}$$

or

$$= 0.872\,664\dots$$

$$\begin{aligned}s &= \theta r \\ &= \left(\frac{5\pi}{18} \right) (35) \quad 1 \text{ mark for substitution} \\ &= \frac{175\pi}{18} \text{ cm} \quad 2 \text{ marks}\end{aligned}$$

or

$$= 30.543 \text{ cm}$$

Question 2

Solve algebraically, where $0 \leq \theta \leq 2\pi$.

$$2 \cos^2 \theta = \sin^2 \theta - 2 \cos \theta$$

Solution

$$2 \cos^2 \theta = 1 - \cos^2 \theta - 2 \cos \theta$$

$$3 \cos^2 \theta + 2 \cos \theta - 1 = 0$$

$$(\cos \theta + 1)(3 \cos \theta - 1) = 0$$

$$\cos \theta = -1 \quad \cos \theta = \frac{1}{3}$$

$$\theta = \pi$$

$$\theta_r = 1.230\,959\dots$$

$$\theta = 1.231, 5.052$$

1 mark for substitution of appropriate identity

1 mark for solving for $\cos \theta$

2 marks for solving for θ (1 mark for each branch)

4 marks

Question 3

Determine the number of arrangements of the letters in the word ATTENTION which begin with the letter A.

Solution

$$\frac{8!}{3!2!} = 3360$$

1 mark for $8!$

1 mark for division by $3!2!$ ($\frac{1}{2}$ mark for $3!$; $\frac{1}{2}$ mark for $2!$)

2 marks

Question 4

Solve for x , algebraically.

$$e^{2x+1} = 5^x$$

Solution

$$\ln(e^{2x+1}) = \ln(5^x) \quad 1/2 \text{ mark for applying logarithms}$$

$$(2x + 1) \ln e = x \ln 5 \quad 1 \text{ mark for power law } (1/2 \text{ mark for each})$$

$$2x + 1 = x \ln 5$$

$$2x - x \ln 5 = -1 \quad 1/2 \text{ mark for collecting terms with } x$$

$$x(2 - \ln 5) = -1$$

$$x = \frac{-1}{2 - \ln 5} \quad \frac{1}{2} \text{ mark for isolating } x$$

$$x = -2.560412\dots$$

$$x = -2.560 \quad 1/2 \text{ mark for evaluating quotient of logarithms}$$

3 marks

Question 5

There are 10 teachers and 17 students who would like to attend a field trip.

Determine the number of ways that 3 teachers and 9 students can be selected given that Mr. Jones and Mrs. Carol, two of the teachers, must be selected to attend the field trip.

Note: A calculator is not required for the remaining test questions.

Solution

$${}_2C_2 \cdot {}_8C_1 \cdot {}_{17}C_9$$
$$194480$$

1 mark for ${}_8C_1$

$\frac{1}{2}$ mark for ${}_{17}C_9$

$\frac{1}{2}$ mark for product of combinations

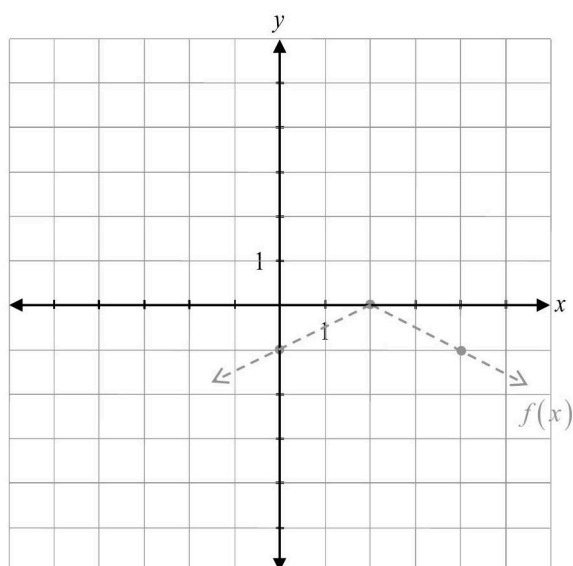
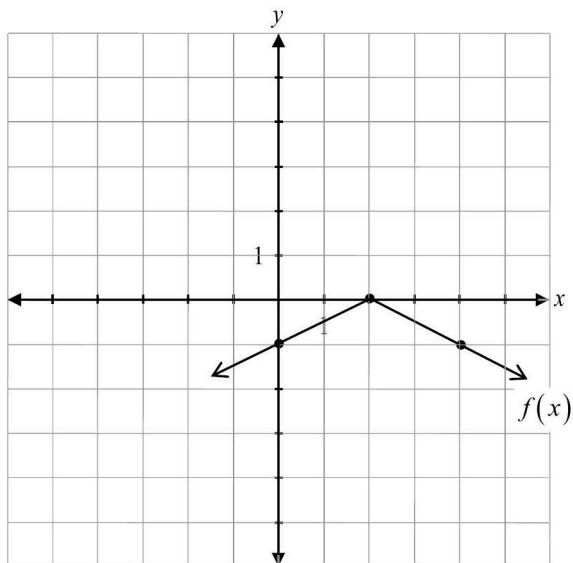
2 marks

Note:

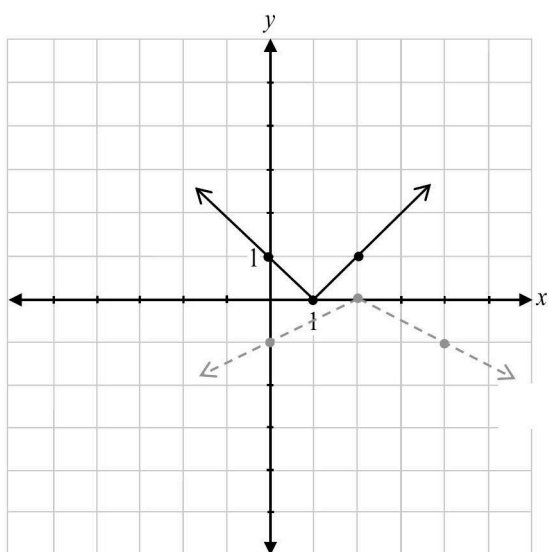
- ${}_2C_2$ does not need to be shown.

Question 6

Given the graph of $f(x)$, sketch the graph of $y = -f(2x)$.



Solution



1 mark for vertical reflection

1 mark for horizontal compression

2 marks

Question 7

There are 5 roads between Anneville and Berrybourg, and 2 roads between Berrybourg and Carriton.

Determine how many ways Blake can travel from Anneville to Carriton and back to Anneville, given the following conditions:

- he must travel through Berrybourg in both directions
- he cannot use the same road twice

Solution

$$\frac{5}{A \rightarrow B} \bullet \frac{2}{B \rightarrow C} \bullet \frac{1}{C \rightarrow B} \bullet \frac{4}{B \rightarrow A}$$

40 ways

$\frac{1}{2}$ mark for correct number of routes between towns

$\frac{1}{2}$ mark for product

1 mark

Question 8

Determine which term contains x^0 in the binomial expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

Solution

Method 1

$x^0 = (x^2)^{6-k} \left(\frac{1}{x}\right)^k$	$\frac{1}{2}$ mark for substitution
$x^0 = x^{12-2k} x^{-k}$	
$x^0 = x^{12-3k}$	
$3k = 12$	
$k = 4$	$\frac{1}{2}$ mark for solving for k
$\therefore$ the fifth term	1 mark for the fifth term (or a term consistent with the value of k)
	2 marks

Method 2

$$\left(x^2\right)^6,\left(x^2\right)^5\left(\frac{1}{x}\right),\left(x^2\right)^4\left(\frac{1}{x}\right)^2$$
$$x^{12}, x^9, x^6 \dots$$

1 mark for determining the pattern

$\therefore$ the fifth term 1 mark for the fifth term (or a term consistent with the pattern)

2 marks

Question 9

Given $f(x) = x^3 + 1$, determine the equation of $f^{-1}(x)$.

Solution

Let $y = f(x)$

$$y = x^3 + 1$$

To determine the inverse of $f(x)$, switch x and y .

$$x = y^3 + 1$$

$$x - 1 = y^3$$

$$\sqrt[3]{x - 1} = y$$

$$f^{-1}(x) = \sqrt[3]{x - 1}$$

1 mark for switching x and y

$\frac{1}{2}$ mark for isolating y

$\frac{1}{2}$ mark for writing equation in terms of $f^{-1}(x)$

2 marks

Question 10

Guillermo was asked to determine the number of ways to select a president, a vice president, and a treasurer from a group of 11 people.

His solution: ${}_{11}C_3$.

Explain why he should have used a permutation instead of a combination.

Solution

He should have used a permutation because the people are being selected for specific positions.

1 mark

Question 11

Prove the following identity for all permissible values of x .

$$\frac{\csc^2 x \sec x}{\tan x + \cot x} = \csc x$$

Left-Hand Side	Right-Hand Side

Solution

Left-Hand Side	Right-Hand Side
$\frac{\left(\frac{1}{\sin^2 x}\right)\left(\frac{1}{\cos x}\right)}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$ $\frac{1}{\frac{\sin^2 x \cos x}{\sin^2 x + \cos^2 x}}$ $\frac{1}{\sin^2 x \cos x} \cdot \frac{\sin x \cos x}{1}$ $\frac{1}{\sin x}$ $\csc x$	$\csc x$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

Determine the value of x , algebraically.

$$5 \log_a 2 - \frac{1}{4} \log_a 16 = \log_a x$$

Solution

$$\log_a 2^5 - \log_a 16^{\frac{1}{4}} = \log_a x \quad \text{1 mark for power law } \left(\frac{1}{2} \text{ mark for each} \right)$$

$$\log_a \left(\frac{32}{2} \right) = \log_a x \quad \text{1 mark for quotient law}$$

$$\log_a 16 = \log_a x$$

$$x = 16 \quad \text{1 mark for equating arguments}$$

3 marks

Question 13

Tamara must determine the factors of $x^4 - 13x + 2x^3 - 14x^2 + 24$.

Explain why the coefficients Tamara used to set up her synthetic division are not written correctly.

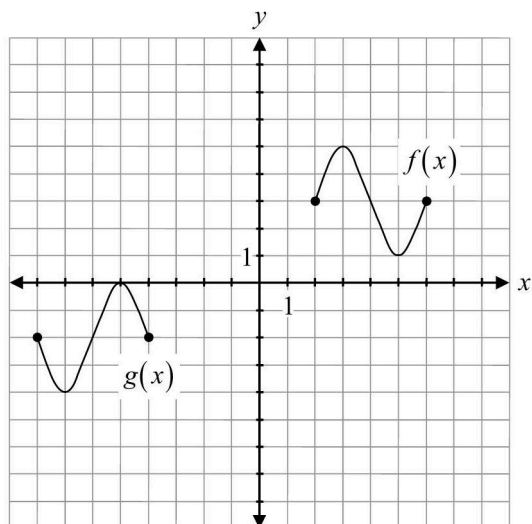
11 -13 2 -14 24

Solution

Tamara did not arrange the coefficients of the terms in order of descending degree.

Question 14

Determine the equation of the graph of $g(x)$ in terms of $f(x)$.



$$g(x) =$$

Solution

$$g(x) = f(-(x + 2)) - 5$$

or

$$g(x) = -f(x + 10) + 1$$

1 mark for horizontal reflection

1 mark for horizontal translation

1 mark for vertical translation

1 mark for vertical reflection

1 mark for horizontal translation

1 mark for vertical translation

3 marks

Question 15

Expand, using the laws of logarithms.

$$\log_2 \left[\frac{(x-1)(x-2)}{x} \right]$$

Solution

$$\log_2(x-1) + \log_2(x-2) - \log_2 x$$

1 mark for product law

1 mark for quotient law

2 marks

Question 16

Identify the range of the function $g(x) = \frac{1}{2}f(x + 1)$, given that the range of the function $y = f(x)$ is $[-6, 4]$.

A.

$[-12, 8]$

B.

$[-7, 3]$

C.

$[-5, 5]$

D.

$[-3, 2]$

Solution

Correct answer: D

$[-3, 2]$

Question 17

Identify the value of a , given that there are 11 terms in the expansion of $(3x^4 - y)^{2a}$.

A.

5

B.

6

C.

10

D.

11

Solution

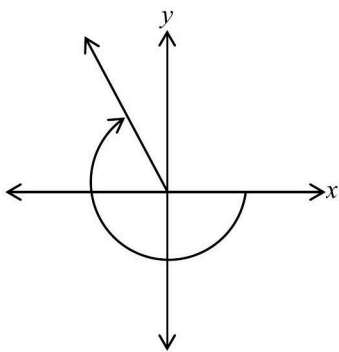
Correct answer: A

5

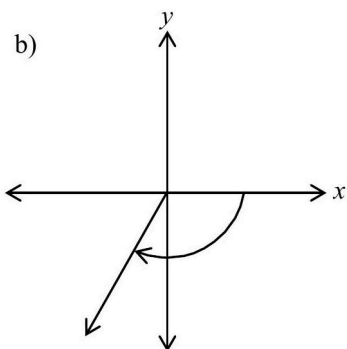
Question 18

Identify the angle that best represents $\theta = -\frac{6\pi}{5}$.

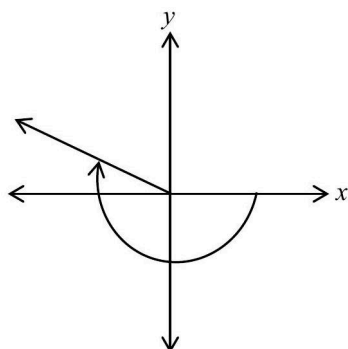
A.



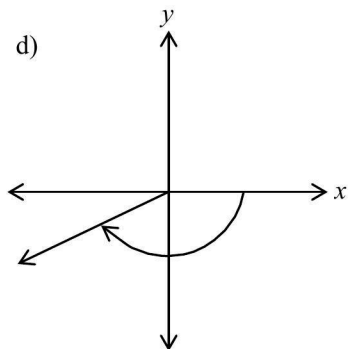
B.



C.

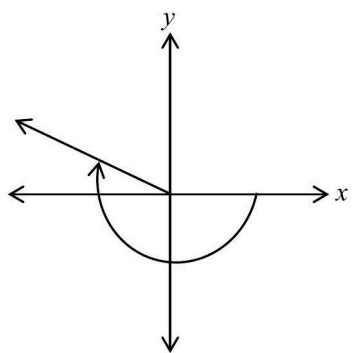


D.



Solution

Correct answer: C



Question 19

Identify a possible value for n , given the graph of $y = -\frac{1}{2}(x+2)^2(x-1)^n$.

A.

1

B.

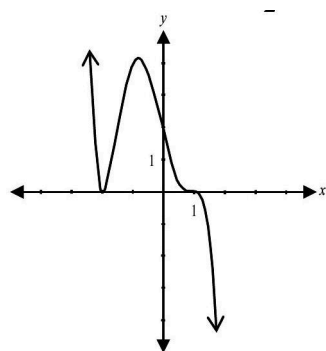
2

C.

3

D.

4



Solution

Correct answer: C

3

Question 20

Identify the statement that is false, given $g(x) = \frac{8x^2}{x^2-16}$.

A.

the graph of $g(x)$ has one x -intercept.

B.

the graph of $g(x)$ has a point of discontinuity (hole) at $x = 0$.

C.

the graph of $g(x)$ has two vertical asymptotes.

D.

the graph of $g(x)$ has a horizontal asymptote at $y = 8$.

Solution

Correct answer: B

the graph of $g(x)$ has a point of discontinuity (hole) at $x = 0$.

Question 21

1 mark

Identify the equivalent form of $\log_a\left(\frac{1}{x^2}\right)$.

A.

$$-2\log_a x$$

B.

$$1 - 2\log_a x$$

C.

$$2\log_a x$$

D.

$$-2\log_a\left(\frac{1}{x}\right)$$

Solution

Correct answer: A

$$-2\log_a x$$

Question 22

Identify which one of the following expressions is equivalent to ${}_{13}C_6$.

A.

$${}_{13}P_6$$

B.

$${}_{13}C_7$$

C.

$${}_{12}P_7$$

D.

$${}_{12}C_6$$

Solution

Correct answer: B

$${}_{13}C_7$$

Question 23

Identify the equation of $h(x) = f(x) - g(x)$, given $f(x) = x + 5$ and $g(x) = 4x + 1$.

A.

$$h(x) = -3x + 6$$

B.

$$h(x) = -3x + 4$$

C.

$$h(x) = 3x + 6$$

D.

$$h(x) = 3x - 4$$

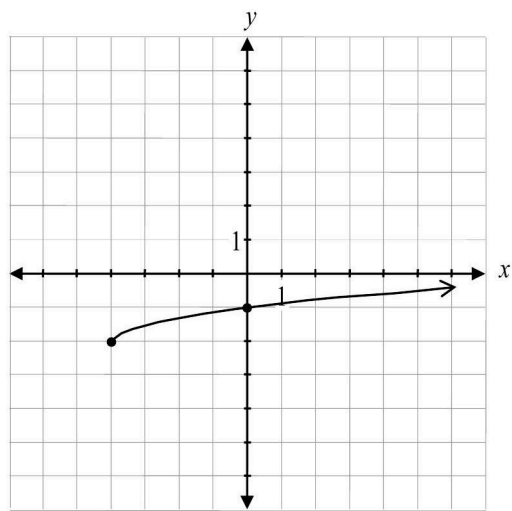
Solution

Correct answer: B

$$h(x) = -3x + 4$$

Question 24

Determine the equation of the radical function represented by the graph.



y =

Solution

$y = \frac{1}{2}\sqrt{x + 4} - 2$	1 mark for vertical compression 1 mark for horizontal translation 1 mark for vertical translation
or	3 marks
$y = \sqrt{\frac{1}{4}(x + 4)} - 2$	1 mark for horizontal stretch 1 mark for horizontal translation 1 mark for vertical translation
	3 marks

Question 25

Determine the exact value of x .

$$\sec\left(\frac{2\pi}{3}\right)\left(\sin\left(-\frac{5\pi}{3}\right)\right)(x) = 3$$

Solution

$$(-2)\left(\frac{\sqrt{3}}{2}\right)(x) = 3$$

$$-\sqrt{3}(x) = 3$$

$$x = -\frac{3}{\sqrt{3}}$$

1 mark for $\sec\left(\frac{2\pi}{3}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

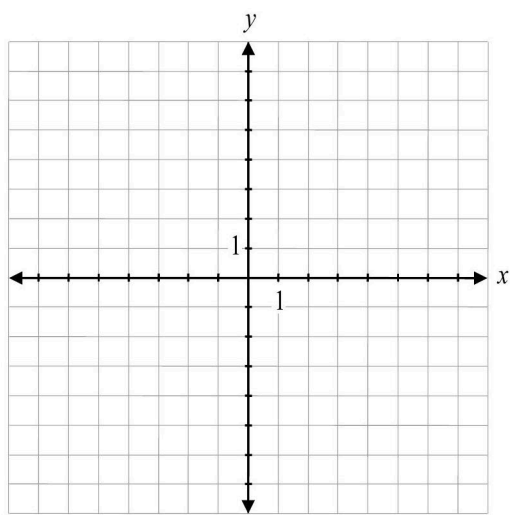
1 mark for $\sin\left(-\frac{5\pi}{3}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

2 marks

$$x = -\sqrt{3}$$

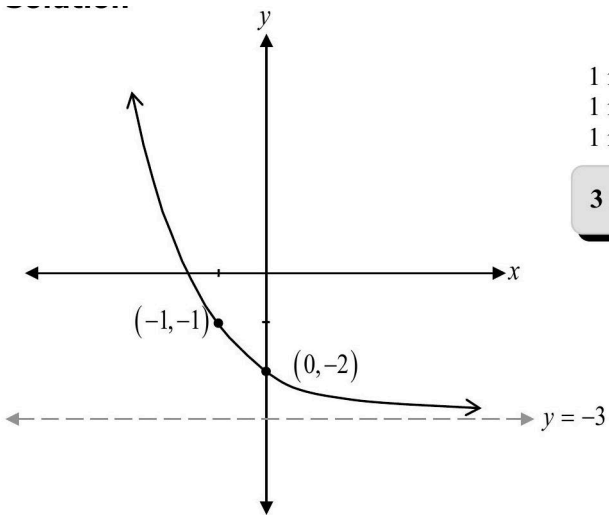
Question 26

Sketch the graph of $y = 2^{-x} - 3$.



Solution

Solution



1 1

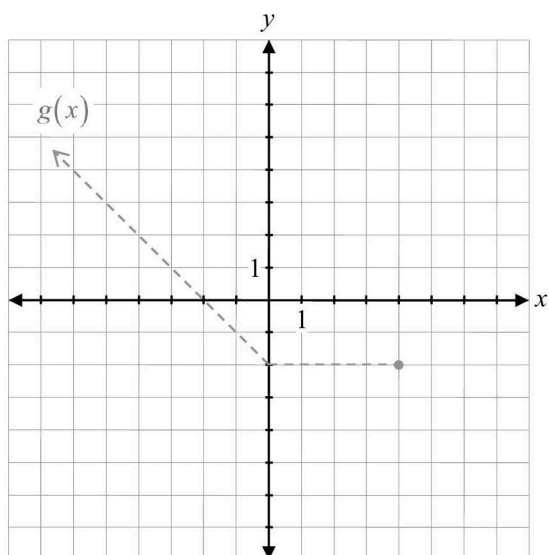
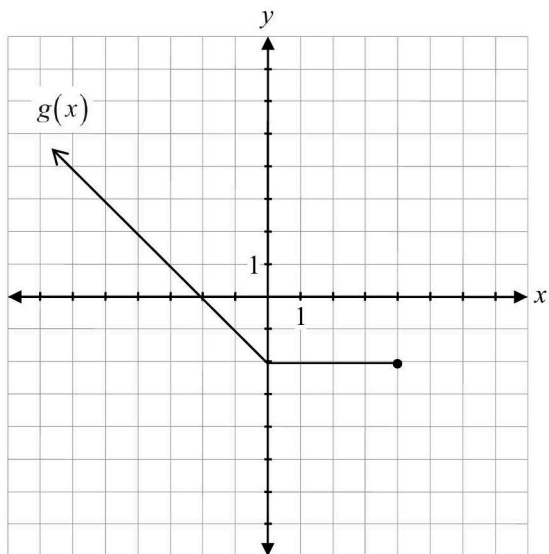
1 1

1 1

3

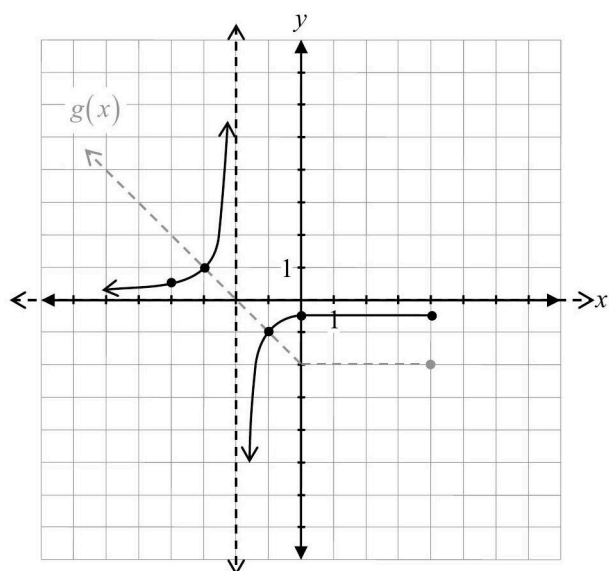
Question 27

Given the graph of $y = g(x)$, sketch the graph of $y = \frac{1}{g(x)}$.



The graph of
 $g(x)$ has already
been drawn for
your reference.
No marks will be
awarded for the
graph of $g(x)$.

Solution



1 mark for asymptotic behaviour approaching

$x = -2$ ($\frac{1}{2}$ mark for each side)

$\frac{1}{2}$ mark for asymptotic behaviour approaching

$y = 0$

$\frac{1}{2}$ mark for graph left of $x = -2$

$\frac{1}{2}$ mark for graph right of $x = -2$

$\frac{1}{2}$ mark for restricting domain

3 marks

Question 28

Determine the exact value of $\tan(\frac{\pi}{12})$.

Solution

$$\begin{aligned}\tan(\frac{\pi}{12}) &= \tan(\frac{4\pi}{12} - \frac{3\pi}{12}) \\ \tan(\frac{\pi}{3} - \frac{\pi}{4}) &= \frac{\tan \frac{\pi}{3} - \tan \frac{\pi}{4}}{1 + \tan \frac{\pi}{3} \tan \frac{\pi}{4}} \\ &= \frac{\sqrt{3}-1}{1+\sqrt{3}} && 1 \text{ mark for substitution into correct identity} \\ &\text{or} && 1 \text{ mark for exact values } (\frac{1}{2} \text{ mark for each value}) \\ &= 2 - \sqrt{3} \\ &\text{or} \\ &= \frac{3-\sqrt{3}}{3+\sqrt{3}}\end{aligned}$$

- Other combinations are possible.

Question 29

Explain why the graph of $g(x) = \frac{3}{x^2+4}$ does not have a vertical asymptote.

Solution

The graph of a rational function has a vertical asymptote when the denominator is equal to zero.

The denominator of $g(x)$ will never be equal to zero.

1 mark

Question 30

Solve, algebraically.

$$\log_3 x + \log_3(x + 8) = 2$$

Solution

Method 1

$$\begin{aligned}\log_3[x(x + 8)] &= 2 \\ x^2 + 8x &= 3^2 \\ x^2 + 8x - 9 &= 0 \\ (x + 9)(x - 1) &= 0 \\ x + 9 \quad x &= 1\end{aligned}$$

1 mark for product law

1 mark for exponential form

$\frac{1}{2}$ mark for the permissible value of x

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Method 2

$$\begin{aligned}\log_3 x + \log_3(x + 8) &= \log_3 3^2 \\ \log_3[x(x + 8)] &= \log_3 9 \\ x^2 + 8x &= 9 \\ x^2 + 8x - 9 &= 0 \\ (x + 9)(x - 1) &= 0 \\ x \geq -9 \quad x &= 1\end{aligned}$$

1 mark for logarithmic form

1 mark for product law

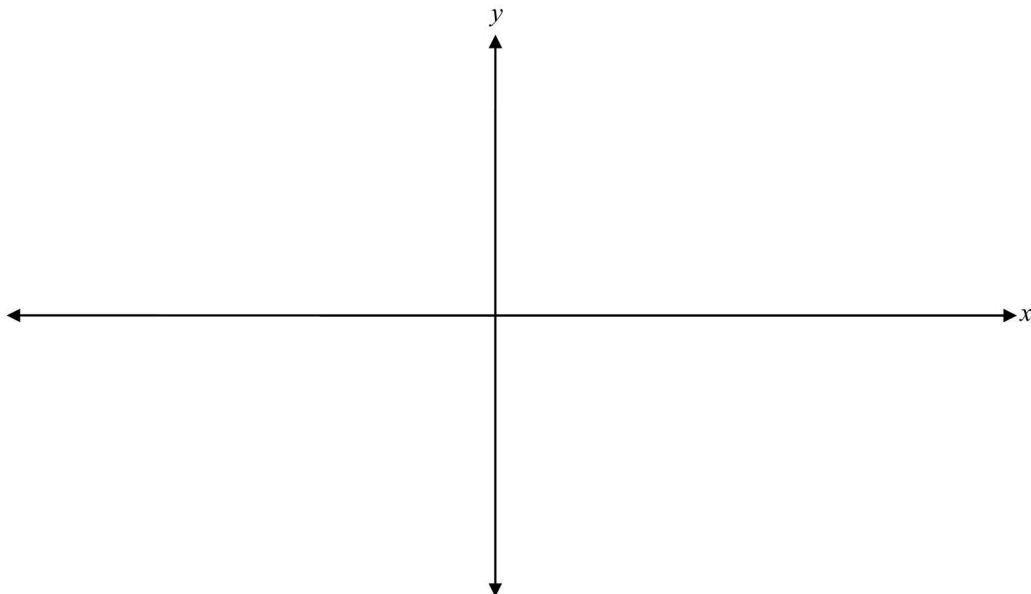
$\frac{1}{2}$ mark for the permissible value of x

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

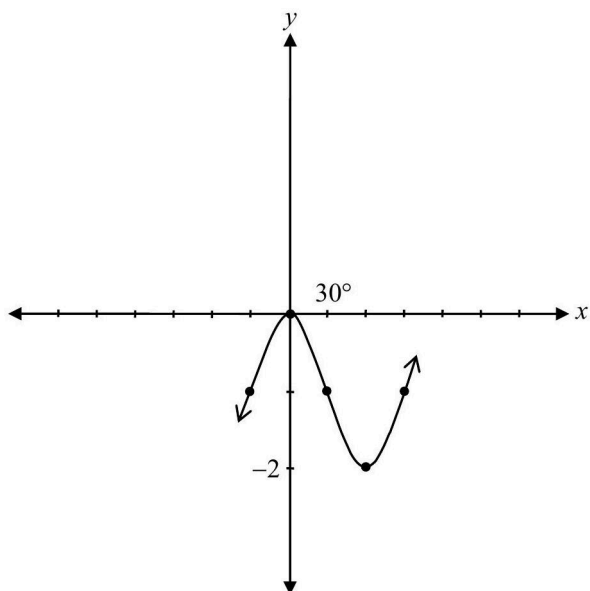
3 marks

Question 31

Sketch at least one period of the graph of the function $y = \sin(3(x + 30^\circ)) - 1$.



Solution



1 mark for shape of a sinusoidal function with correct amplitude

1 mark for period

1 mark for horizontal translation

1 mark for vertical translation

4 marks

Question 32

Explain why the domain of the function, $f(x) = \log(x - 3)$, is $x > 3$.

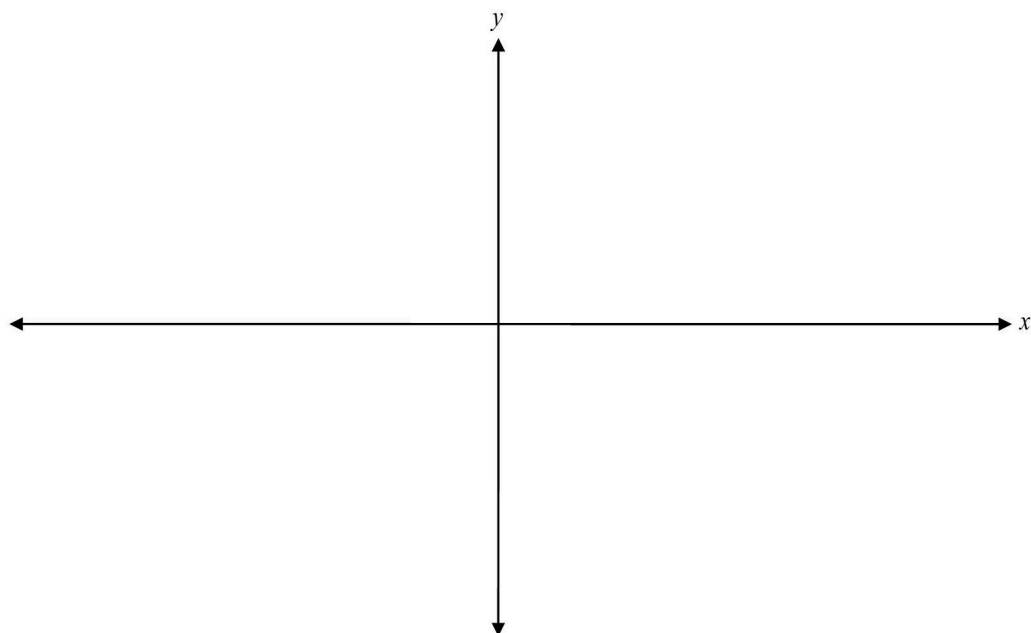
Solution

The argument of a logarithm cannot be negative or zero.

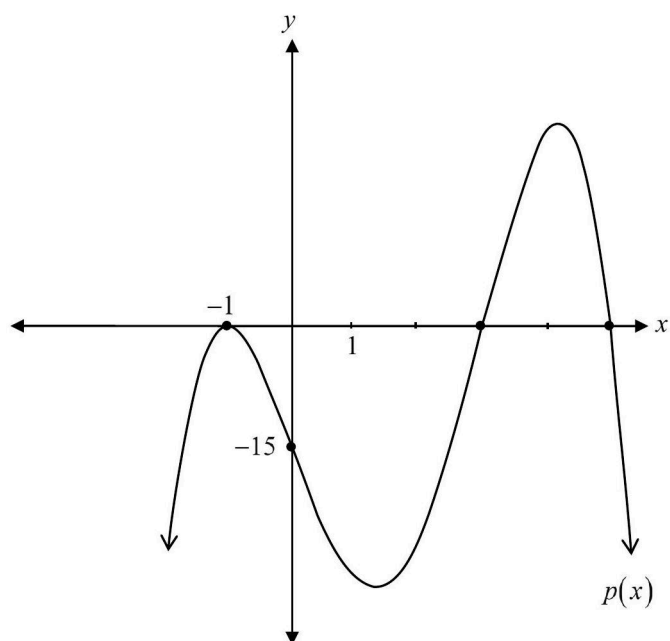
1 mark

Question 33

Sketch the graph of $p(x) = -(x-3)(x+1)^2(x-5)$.



Solution



1 mark for x -intercepts

1 mark for multiplicity of 2 at $x = -1$

$\frac{1}{2}$ mark for end behaviour

$\frac{1}{2}$ mark for y -intercept

3 marks

Question 34

Given that $\sin \theta = -\frac{2}{3}$ and $\tan \theta > 0$, determine the exact value of $\sin 2\theta$.

Solution

$$x^2 = r^2 - y^2$$

$$x^2 = 9 - 4$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$$x = -\sqrt{5}$$

$$\cos \theta = -\frac{\sqrt{5}}{3}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2\left(-\frac{2}{3}\right)\left(-\frac{\sqrt{5}}{3}\right)$$

$$= \frac{4\sqrt{5}}{9}$$

$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of $\cos \theta$

2 marks

Note:

- Accept any of the following values for x : $x = \pm\sqrt{5}$, $x = -\sqrt{5}$, or $x = \sqrt{5}$.

Question 35

Justify whether $\frac{5\pi}{8}$ and $-\frac{11\pi}{4}$ are coterminal angles.

Solution

Method 1

$$\frac{5\pi}{8} - \left(-\frac{11\pi}{4}\right)$$

$$\frac{5\pi}{8} + \frac{11\pi}{4}$$

$$\frac{5\pi}{8} + \frac{22\pi}{8}$$

$$\frac{27\pi}{8}$$

Since $\frac{27\pi}{8}$ is not a multiple of 2π they are not coterminal angles.

1 mark

Method 2

Coterminal angles of $\frac{5\pi}{8}$:

$$\frac{5\pi}{8} - \frac{16\pi}{8} = -\frac{11\pi}{8}$$

$$-\frac{11\pi}{8} - \frac{16\pi}{8} = -\frac{27\pi}{8}$$

Coterminal angles of $-\frac{11\pi}{4}$:

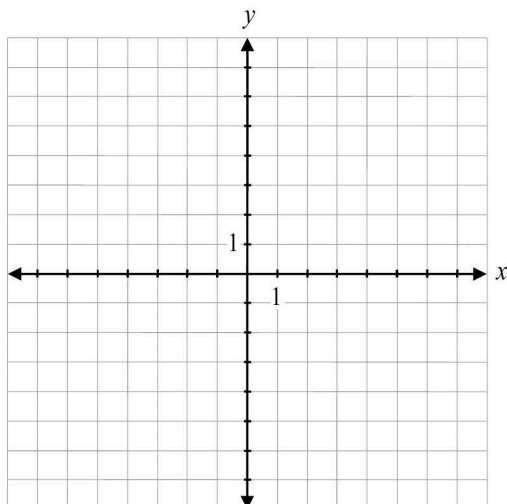
$$-\frac{11\pi}{4} \left(\frac{2}{2}\right) = -\frac{22\pi}{8}$$

$\therefore$ Angles are not coterminal since $-\frac{27\pi}{8} \neq -\frac{22\pi}{8}$.

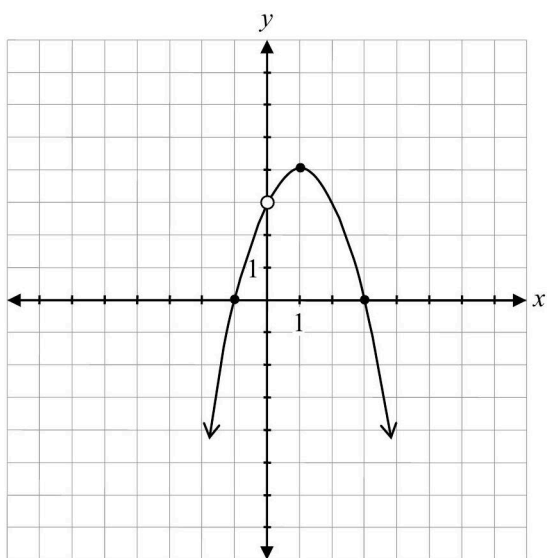
1 mark

Question 36

Sketch the graph of $f(x) = \frac{-2x(x+1)(x-3)}{2x}$.



Solution



1 mark for point of discontinuity (hole) at $x = 0$

$\frac{1}{2}$ mark for shape of a parabola with correct
 x -intercepts

$\frac{1}{2}$ mark for end behaviour

2 marks

Note:

- Deduct $\frac{1}{2}$ mark for procedural error (incorrect y -value for point of discontinuity (hole)).

Question 37

Given $\frac{\sin \theta + \cos \theta \csc \theta}{\sin \theta}$, determine the non-permissible values of θ , where $\theta \in \mathbb{R}$.

Solution

Method 1

$\sin \theta = 0$	$\frac{1}{2}$ mark for $\sin \theta = 0$
$\theta = 0, \pi, 2\pi$	$\frac{1}{2}$ mark for any non-permissible value of θ
$\theta = \pi k, k \in \mathbb{Z}$	1 mark for all non-permissible values of θ

2 marks

Method 2

$\sin \theta = 0$	$\frac{1}{2}$ mark for $\sin \theta = 0$
$\theta = 0, \pi, 2\pi$	$\frac{1}{2}$ mark for any non-permissible value of θ
$\left. \begin{array}{l} \theta = 2\pi k, k \in \mathbb{Z} \\ \theta = \pi + 2\pi k, k \in \mathbb{Z} \end{array} \right\}$	1 mark for all non-permissible values of θ
	2 marks

Question 38

Write an equation of a rational function that has a horizontal asymptote at $y = 0$ and a vertical asymptote at $x = 6$.

Solution

$f(x) = \frac{1}{x-6}$ 1 mark for horizontal asymptote

1 mark for vertical asymptote

2 marks

Note:

- Other answers are possible.

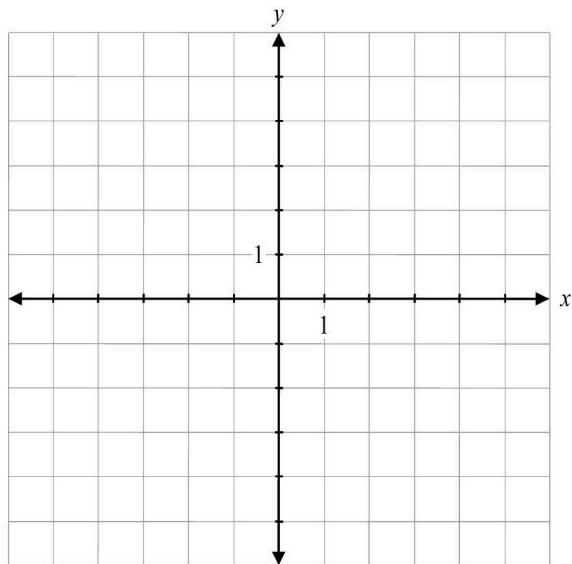
Question 39

Given the functions $f(x) = \sqrt{x-1}$ and $g(x) = x^2$,

a) state the equation of $g(f(x))$.

$$g(f(x)) =$$

b) sketch the graph of $g(f(x))$.

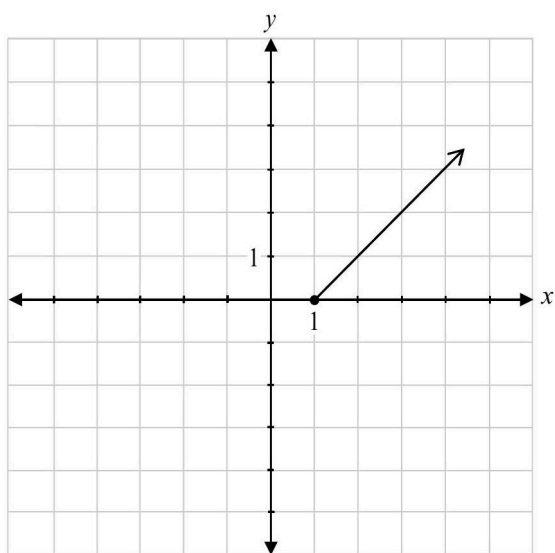


Solution

a) $g(f(x)) = x - 1, x \geq 1$

1 mark

b)



1 mark for shape of graph consistent with a)

1 mark for restricted domain

2 marks

Notes:

- Deduct a maximum of 1 mark for the concept error of not restricting the domain.
- Deduct $\frac{1}{2}$ mark for procedural error (not stating domain) if graph shows correct domain.

Question 40

Suzanne was asked to determine the value of $\tan \theta$, given that $\sec \theta = -\frac{8}{3}$ and θ terminates in quadrant II.

Her solution:

$$(-3)^2 + y^2 = (8)^2$$

$$y^2 = 55$$

$$y = \sqrt{55}$$

$$\tan \theta = \frac{\sqrt{55}}{3}$$

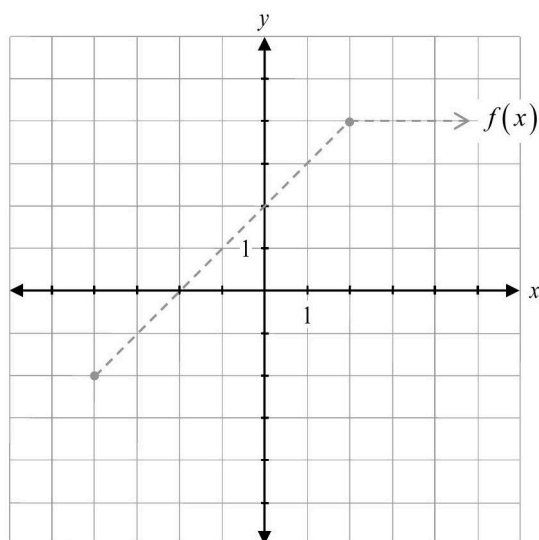
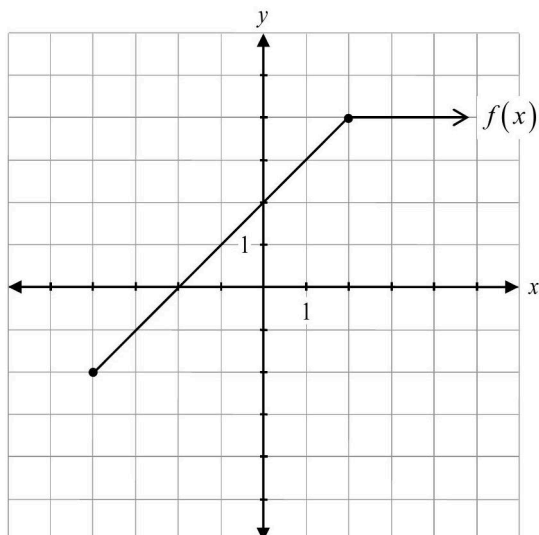
Describe her error.

Solution

Suzanne did not consider that the value of $\tan \theta$ is negative in quadrant II.

Question 41

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



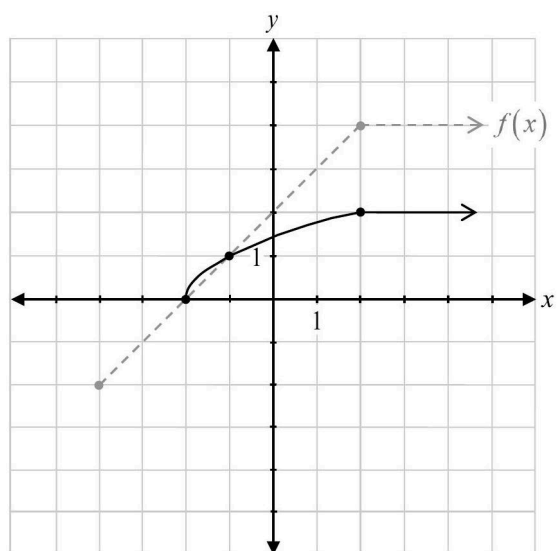
The graph of



$f(x)$ has already
been drawn for
your reference.

No marks will be
awarded for the
graph of $f(x)$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points

$\frac{1}{2}$ mark for shape to the right of invariant points

2 marks

Question 42

The point $P(\theta) = (0, -1)$ lies on the unit circle. State the angle θ , over the interval $[2\pi, 4\pi]$.

Solution

$$\theta = \frac{7\pi}{2}$$

1 mark

Question 43

Describe how the transformations of $f(x)$ on the graphs of $g(x) = f(3x - 6)$ and $h(x) = f(3(x - 6))$ are different.

Solution

The graph of $g(x)$ is a horizontal translation of $f(x)$ two units to the right and the graph of $h(x)$ is a horizontal translation of $f(x)$ six units to the right.

1 mark

Question 44

a) Solve.

$$\sqrt{2x+5}-3=0$$

b) Describe how the solution in a) relates to the graph of $y = \sqrt{2x+5}-3$.

Solution

a)

$$(\sqrt{2x+5})^2 = 3^2$$

$$2x+5=9$$

$$2x=4$$

$$x=2$$

1 mark

b) The solution is the x -intercept of the graph.

1 mark

Question 45

Determine all of the zeros of the function $p(x) = x^3 - 2x^2 - 9x + 18$.

Solution

$$p(2) = 2^3 - 2(2)^2 - 9(2) + 18 \quad 1 \text{ mark for identifying one possible zero of } p(x)$$
$$= 0$$
$$\therefore (x - 2) \text{ is a factor of } p(x)$$

2	1	-2	-9	18
		2	0	-18
	1	0	-9	0

1 mark for synthetic division (or equivalent strategy)

$$(x - 2)(x^2 - 9) = 0 \quad 1/2 \text{ mark for consistent factors}$$
$$(x - 2)(x - 3)(x + 3) = 0$$
$$x = 2, x = 3, x = -3 \quad 1/2 \text{ mark for consistent zeros}$$

3 marks

Question 46

Given that the point $\left(\frac{\sqrt{23}}{6}, y\right)$ is on the unit circle, determine the exact value(s) of y .

Solution

$$\left(\frac{\sqrt{23}}{6}\right)^2 + y^2 = 1 \quad \frac{1}{2} \text{ mark for substitution}$$

$$\frac{23}{36} + y^2 = 1$$

$$y^2 = \frac{36}{36} - \frac{23}{36}$$

$$y = \pm \frac{\sqrt{13}}{6} \quad 1/2 \text{ mark for exact values of } y$$

1 mark

Question 47

State one zero of the function $y = \tan x$.

Solution

Any solution where $x = \pi k, k \in \mathbb{Z}$, is acceptable.